OSM / Opto-PING Rung 0J

Preregistered Synthetic Lock-In and External Reproducibility Pack v0.1

Executive verdict: synthetic lock-in pass for the primary composite endpoint K.

This gate froze the model equations, parameter grids, perturbation order, K-lock criteria, failure thresholds, random-seed policy, alternative models, and reporting template before running a fresh seed set.

Locked spec SHA-256:

```
d93f6146e4e77ffeffbdd5c133250fd6d474317d8af6c34c3354dd6b28756282
```

Locked design

Primary estimand: K = sqrt(eta_E * zeta_E). Directional eta_E and zeta_E remain secondary conditional endpoints.

Item	Value
eta_E true	4.000
zeta_E true	180.000
K true	26.832816
Fresh seeds	910001-910060
Trials	60

Primary results

Metric	Result
K within 15% rate	1.000
K within 10% rate	1.000
Median K percent error	2.82%
Mean K percent error	3.10%
Full reciprocal CV fold win rate	0.872
Per-trial full-CV >=80% rate	0.833
Directional eta/zeta <=20% rate	1.000
Directional eta/zeta <=15% rate	1.000

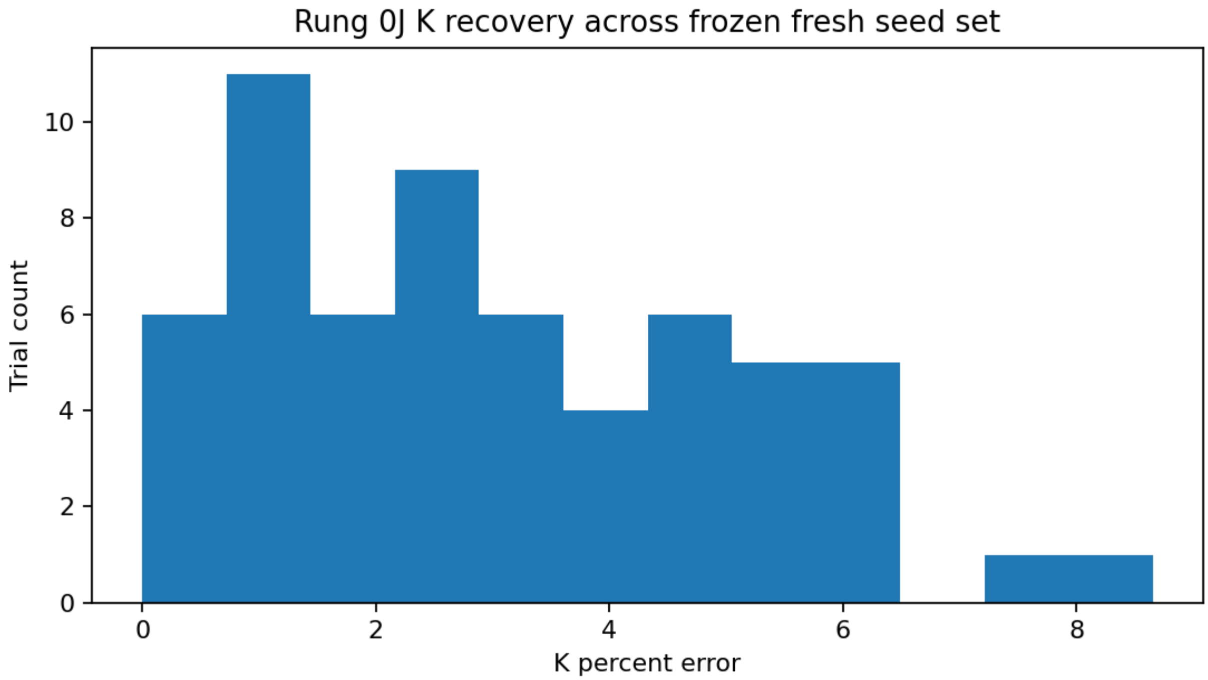
Lock decisions

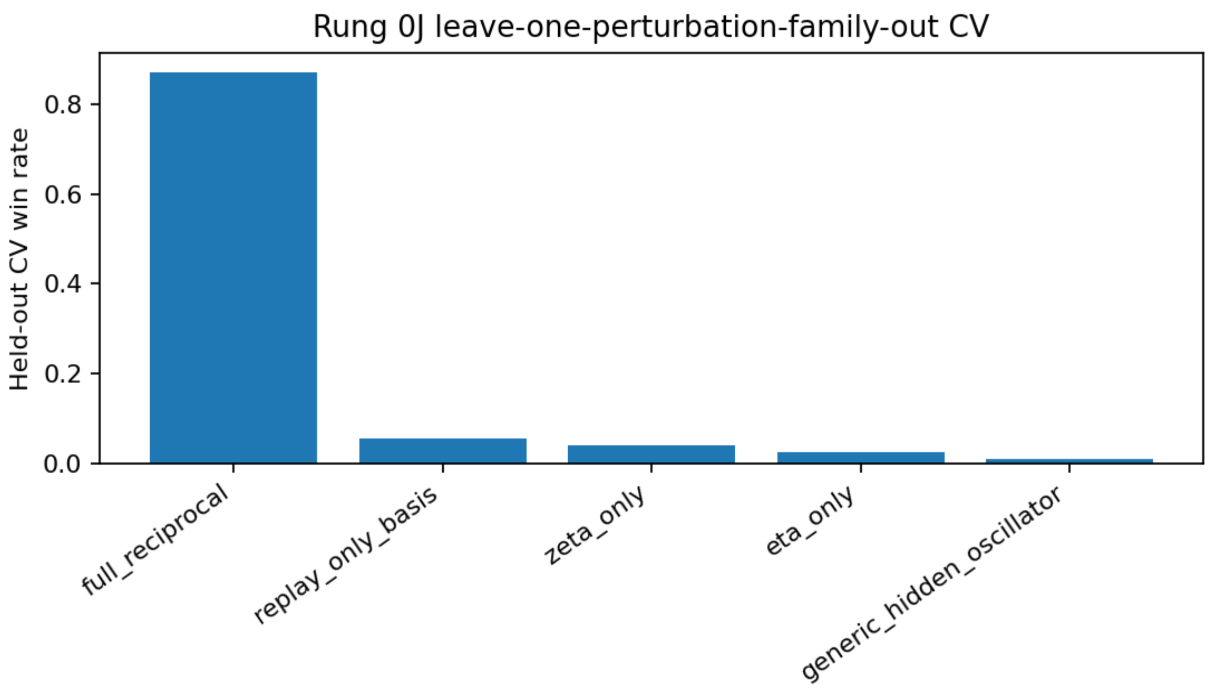
Criterion	Observed	Threshold	Pass
Primary K-lock power	1.0	>=0.90 trials K error <=15%	YES
Median K precision	0.0282	<=0.10 median K error	YES
Strict K-lock secondary	1.0	>=0.80 trials K error <=10%	YES
Held-out specificity	0.8722	>=0.80 full reciprocal CV fold win rate	YES
Directional secondary	1.0	>=0.80 trials eta and zeta each <=20% error	YES
Alternative dominance warning	0.0556	<=0.20 for any single non-full alternative	YES

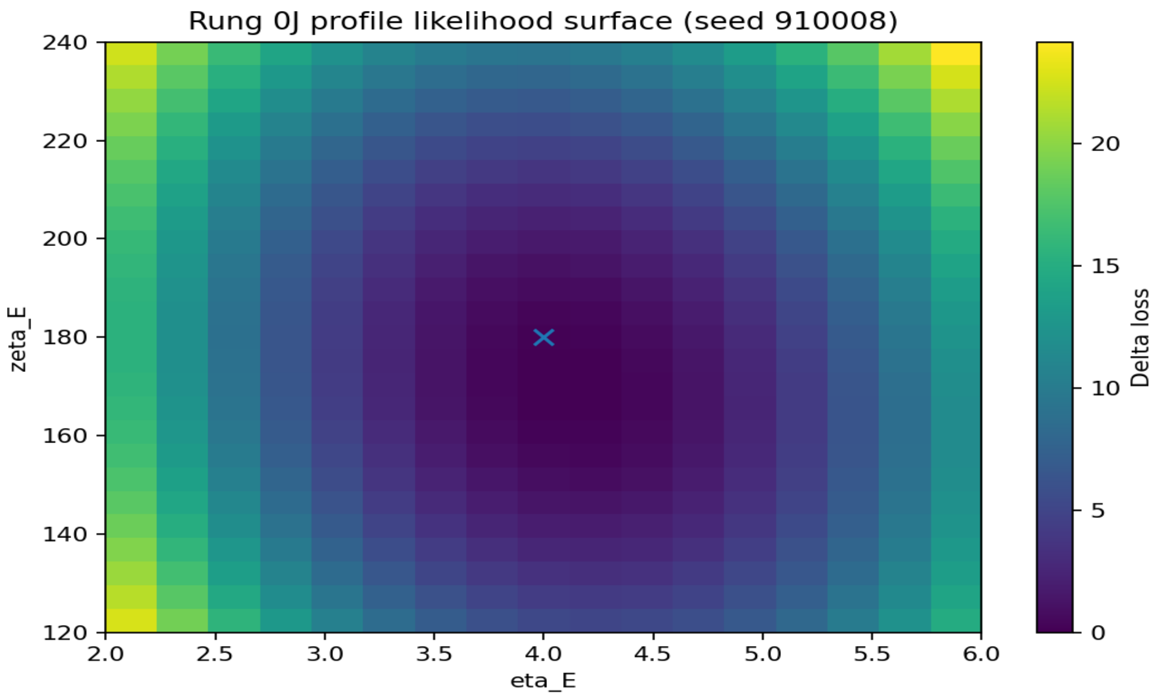
Held-out model comparison

Model	Wins	Win rate
full_reciprocal	314	0.872
replay_only_basis	20	0.056
zeta_only	14	0.039
eta_only	9	0.025
generic_hidden_oscillator	3	0.008

Figures


Rung 0J K recovery across frozen fresh seed set
Trial count
K percent error
0
2
4
6
8
10


Rung 0J leave-one-perturbation-family-out CV
Held-out CV win rate
0.0
0.2
0.4
0.6
0.8
full_reciprocal
replay_only_basis
zeta_only
eta_only
generic_hidden_oscillator



Profile-likelihood ridge summary

Representative seed: 910008. Near-cell count: 31. Eta width: 0.750. Zeta width: 40.000. K width: 4.358.

Interpretation: Profile geometry is constrained enough for K-lock in this synthetic gate. Eta/zeta separation remains secondary and conditional because earlier stress gates showed ridge-confounding under harder hidden-Y and proxy-imperfection regimes.

Boundary

Synthetic identifiability only. This report does not prove OSM, microtubular memory, LTP, dendritic spine growth, quantum memory, or a human EEG mechanism.